

Jayden (Jiafeng) Yu

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TECHNICAL SKILLS

Languages: Python, C/C++, Rust, JavaScript/TypeScript, KDB/SQL, Java/Kotlin, Lisp, Bash, Lua, MIPS/ARM, R
Tools: PyTorch, TensorFlow, NumPy, Pandas, Node.js, Vue.js, MongoDB, Redis, Git, Linux/FreeBSD, LLVM, AWS
Interested in: OCaml, Automated Theorem Proving, Quantum Computing, Pen.Testing, Gentoo/BlackArch Linux
Communication: English, French, Mandarin Chinese, Spanish

EXPERIENCE

OncAI San Francisco, California
Machine Learning Engineer Intern | Python Jan 2024 - May 2024

- Engineered a **3D UNet Diffusion Model** to render synthetic CT/PET scans for medical **data generation**
- Researched and implemented various **data augmentation algorithms** to enhance **learning robustness**
- Implemented and updated features to the **Valohai MLOps pipeline** to unify interfaces
- Implemented **Python** scripts to automate data extraction from medical records

University of Waterloo Waterloo, Ontario
Undergraduate Research Assistant | Python Oct 2023 - Ongoing

- Researching **automated proof generation** using **LLM** and performing **proof verification** with **Coq**

Ledn Inc Toronto, Ontario
Quantitative Developer Intern | Rust, Python Sept 2022 - Dec 2022

- Designed, engineered, and implemented from scratch a **Rust** based multi-threaded market data consolidator to decrease processing latency **by over 50%** when compared to a FOSS Python model.
- Engineered the design and implementation of a **ML** quant trading product in **Python** and **Rust**, through researching various limit order models, efficient documentation, and code reviews
- Fine-tuned crypto market data to produce supervised learning datasets using **KDB** and **SQL**
- Managed an **AWS EC2** instance to source, store, and update modeling data and host a KX Dev environment
- Researched, implemented, and performance tested a **LSTM** model in **PyTorch** for price prediction

Ford Motor Company Waterloo, Ontario
Software Engineer Intern | Python, JavaScript Jan 2022 - April 2022

- Implemented the **backend development** of a performance metrics website in **JavaScript**
- Engineered **Python** scripts to automate updates to the **data processing pipeline** via **JSON** configuration files

SS&C Technologies Toronto, Ontario
Full Stack Developer Intern | Java, JavaScript May 2021 - Sept 2021

- Implemented modules of a **Java** based **Selenium** webpage testing framework from scratch
- Designed and implemented **frontend** components of a revamped legacy product using **Vue.js**

EDUCATION

University of Waterloo Waterloo, ON
Bachelor of Mathematics in Honours Computer Science, minor in Pure Mathematics Sep 2020 - May 2025E

- Relevant Coursework:** ML, AI, Computer Vision, Algorithms, Data Structures, OS, OOP, Numerical Computation, Compiler Construction, Formal Methods, Computer Architecture, Abstract Algebra, Real Analysis

COMMUNITY

University of Waterloo Computer Science Club Waterloo, ON
Systems Administrator Mar 2023 - Aug 2023

- Managed CS Club services and helped maintained uWaterloo's tier 1 **Arch Linux** mirrors
- Helped allocate server space and host websites for other student clubs

PROJECTS

Immune System Invaders | *JavaScript, HTML, CSS* Mar 2020
Created a web based implementation of the Space Invaders game with a COVID-19 twist in standard **JavaScript, HTML, CSS** with **OOP** principles, scalability, and maintainability in mind